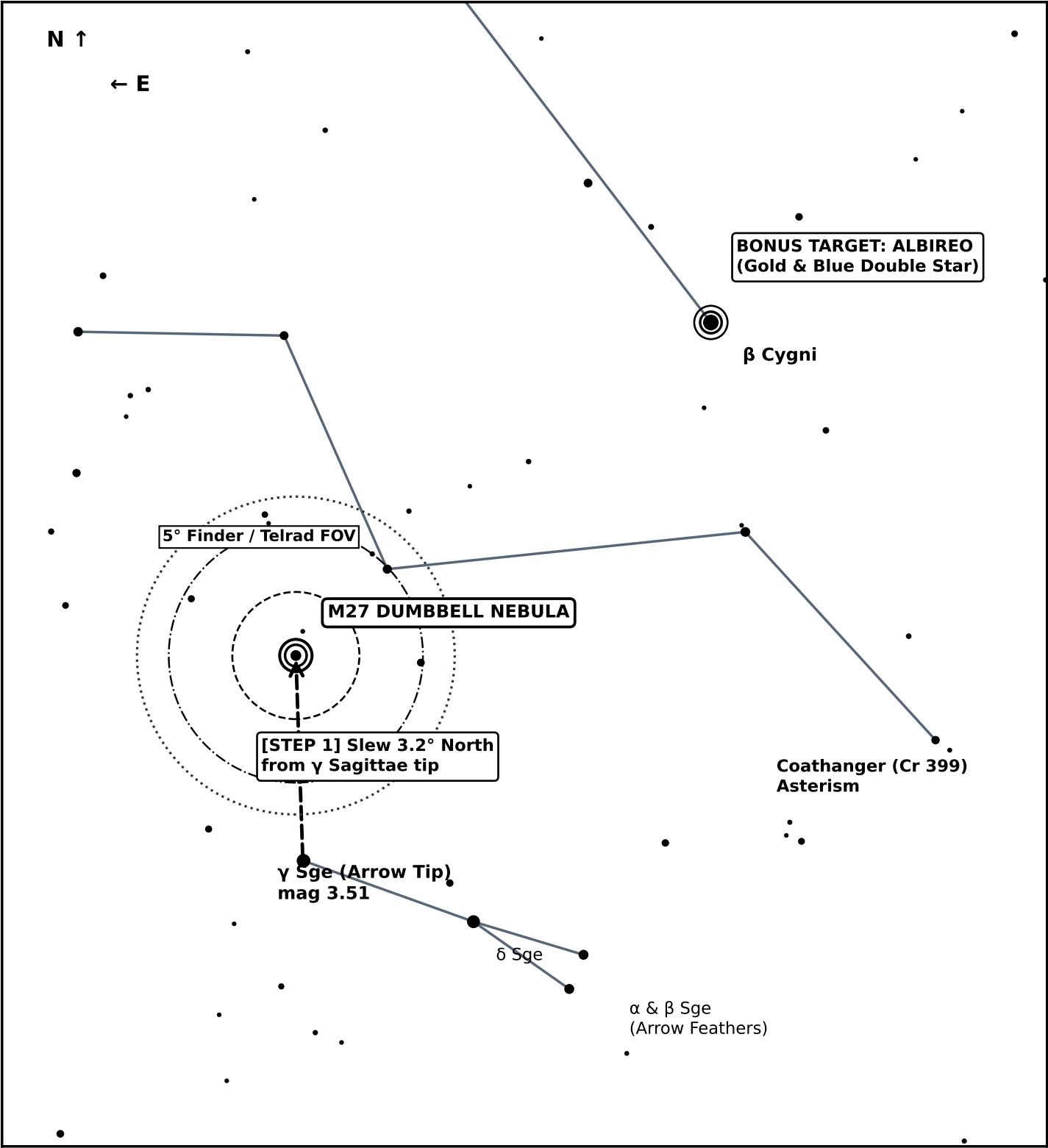


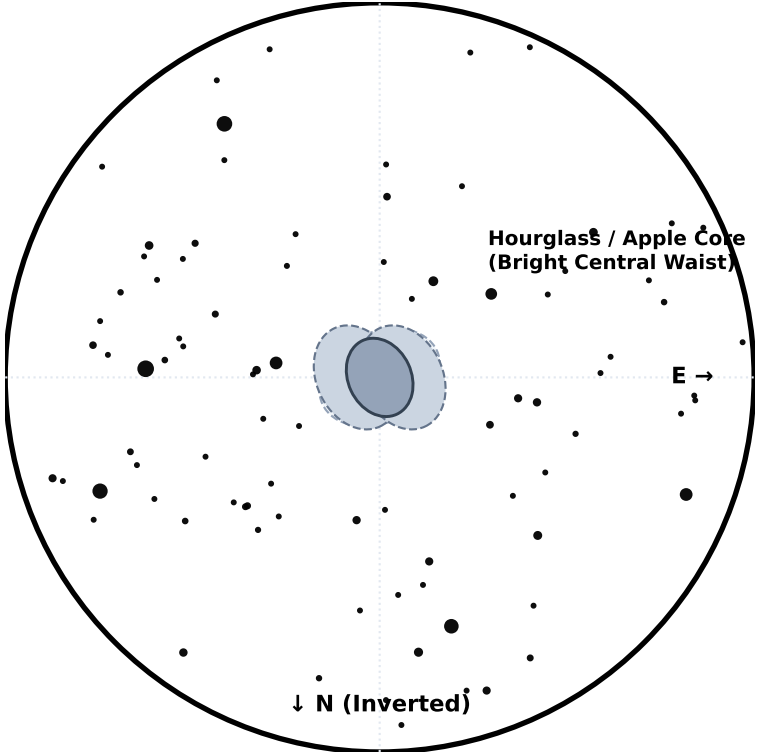
STAR-HOPPING FINDER CHART 2: M27 (DUMBBELL NEBULA) & ALBIREO — VULPECULA / CYGNUS

Location: Dvigrad (Bortle 4, SQM 21.0) | Sep 12, 2026 • 21:15 CEST | Sky-Watcher 200P Dobsonian (8" f/6, 1200mm FL) | B/W Toner-Saver Edition

VIEWPORT A: WIDE-FIELD STAR-HOPPING CHART (Upright Naked-Eye / Finder Orientation: N ↑, E ←)



VIEWPORT B: TELESCOPE EYEPIECE SIMULATION (Negative Inverted 180°: N ↓, E →)



20mm Eyepiece (60× Magnification, 50' True FOV)

VIEWPORT C: TARGET DOSSIER & STAR-HOPPING INSTRUCTIONS

OBJECT DOSSIER: M27 (NGC 6853) & ALBIREO

- Constellation: Vulpecula | Type: Planetary Nebula
- Magnitude: 7.5V | Surface Brightness: 11.2 mag/arcmin²
- Apparent Size: 8.0' × 5.6' | Distance: ~1,360 light-years
- Tonight's Ephemeris at Dvigrad (Sep 12, 2026):
 - 20:00 CEST: Alt 60.0°, Az 147° (Southeast)
 - 21:15 CEST: Alt 67.2°, Az 166° (Meridian Sweet Spot!)
 - 22:30 CEST: Alt 65.3°, Az 189° (High & Pristine)

OBSERVING STRATEGY & EYEPIECES:

- Finder (20mm / 60×): Ideal magnification! Frames the entire classic 'apple-core' dumbbell glow against dark space.
- Detail (12.5mm / 96×): Reveals the brighter outer rims and faint extensions connecting the two main lobes.
- Albireo (β Cyg): Split the 34.3" pair into radiant topaz-gold (mag 3.1) & sapphire-blue (mag 5.1)!

STEP-BY-STEP STAR HOP NARRATIVE:

1. Naked-Eye: Locate Sagitta the Arrow between Altair and Albireo. Identify tip star γ Sagittae (mag 3.5).
2. Telrad Hop: Place γ Sagittae on the southern outer 4° ring; the central reticle points directly at M27 (3.2° North).
3. Eyepiece confirmation: A striking, bright ethereal hourglass cloud appears instantly in the 20mm field!